Supplementary Information (Online Resource 7)

Article title: How Teams Decide on Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Qualitative evaluation results

The qualitative research included interviews with five experts in human factors engineering. Table 1 outlines the demographic profiles of the experts interviewed.

Participant #	Expertise	Years of experience	Domain and context
E1	Expert and consultant in human-machine interaction	>20	Military
E2	Human-machine interaction designer in systems that include AI components	>10	Safety critical
E3	Expert in Human Factors Engineering with a strong background in systems engineering, human-machine interaction, and autonomous systems.	>20	Autonomous systems, robotics
E4	AI system development expert	>10	Public-sector safety, surveillance
E5	Data science expert	>20	Military, autonomous systems, swarms

Table 1 – Participants demography

During these interviews, we presented the theory and invited the experts to evaluate it and suggest possible improvements. To facilitate richer feedback, we also prepared a set of complementary questions designed to prompt further reflection and critique, including:

1 – How effectively does the initial phase of this framework support capturing real-world operational context and user diversity? What additional contextual elements (e.g., cognitive workload, environmental stressors) should be included?

2 – Does the framework capture all relevant oversight dimensions (e.g., control, transparency, trust, intervention readiness)?

3 – What user-centered design principles should guide the implementation of oversight mechanisms?

4 – What evaluation methods (e.g., simulation, cognitive walkthroughs, usability testing) are most effective for refining oversight mechanisms?

5 – Have you used structured methods (e.g., MCDA, WWW models, workshops) to resolve trade-offs in past projects?

We conducted the deductive thematic analysis [Proudfoot], using Atlas.ti, on the interviews transcripts to identify key themes corresponding to each stage of the framework. The following sections present the main themes extracted from the interviews, organized by framework stages. We marked each theme extracted from the interviews as fully or partially covered in the original framework or as a new theme that needs to be added to the framework.

Legends:

- **Theme** – A key insight or recurring idea derived from expert interviews.
- **Coverage** – Indicates whether and how the theme is currently addressed in the existing framework. F - fully represented, P - partially represented, or N - absent as a distinct element
- **New** – Highlights a novel decision point or activity suggested by the theme that should be incorporated into the framework.
- **Experts** – Reports the identifiers of distinct experts whose available source material supports the theme.

- **Records** – Reports the number of distinct supporting passages identified for the theme in the available interview transcripts or interview summaries. Multiple passages may come from the same expert, so the number of records may exceed the number of experts.

I. PHASE 1 - DEFINE SYSTEM CONTEXT & CONSTRAINTS

Theme	Coverage	New	Experts	Records
P1.1 User Role Differentiation	Partial	+	E1, E2, E3, E5	10
P1.2 Domain and Risk Assessment	Full		E2, E3, E4, E5	7
P1.3 Ethical and Regulatory Constraints	Full		E3, E4, E5	3
P1.4 Trust and Perception	Partial	+	E1, E2, E3, E5	7
P1.5 Environmental constraints	Partial	+	E1, E2	2
P1.6 Domain specificity	Full		E1, E2, E3, E5	9
P1.7 Risk factors	Full		E2, E3, E4	3
P1.8 Perceived responsibility and accountability	Partial	+	E3	3
P1.9 User motivation	None	+	E3	1

The following provides insight into new themes uncovered during the interviews:

P1.1 Differentiation of User Roles: It is essential to distinguish between types of users, such as service recipients, system operators, and proactive users, as each role entails different oversight requirements. A one-size-fits-all approach to oversight is insufficient; involvement must be tailored to the user's role and context. E1 Transcript extract – "*It very much depends on the role of the person you want to involve with the system: whether they use the system or receive a service from it.*"

P1.4 Trust and Perception: Users' willingness to engage with the system is strongly shaped by psychological factors, including trust, prior experience, and perceived system reliability. These elements directly impact on how much oversight users are comfortable exercising. E2 Transcript extract – "*It doesn't matter how reliable the system actually is, what matters is how reliable the user perceives it to be.*"

P1.5 Environmental Constraints: The operational setting, such as high-stress or high-tempo environments like battlefields or traffic control centers, can limit the feasibility of active oversight and situational awareness, necessitating adaptive system support. E2 Transcript extract – " *In battlefield or operational situations there are influences that are very difficult to predict in the laboratory*."

P1.8 Perceived responsibility and Accountability: Effective oversight relies not only on available tools, but also on whether users have or feel a sense of responsibility for system outcomes. When accountability is unclear or diminished, users may become passive and overly reliant on automation, even when intervention is possible. E3 Transcript extract – "*If we want the person to intervene, they have to understand what their responsibility is*."

P1.9 User Motivation: High levels of intrinsic or extrinsic motivation (e.g., ethical responsibility, mission urgency) can drive users to maintain engagement with the system despite usability challenges. Motivation thus acts as a critical enabler of sustained oversight behavior. E3 Transcript extract – "*What we saw with our military operators was that, regardless of how much we made things difficult for them, they succeeded, because motivation was a factor."*

II. PHASE 2 - ESTABLISH OVERSIGHT SPECTRUM

Theme	Coverage	New	Experts identifiers	Records
P2.1 Clarifying levels of automation	Partial	+	E1, E3	7

P2.1 Clarifying Levels of Automation (LOA): The traditional HITL/HOTL/HOOTL taxonomy provides an overly simplistic view of human roles in complex systems. There is a growing need for more precise definitions and transparent communication of automation levels, enabling users to clearly understand their expected involvement and responsibilities within the system. E3 Transcript extract – "*The intermediate positions could amount to a hundred possibilities. Once you talk about which routines are automated, the main question is what those levels actually are.*"

III. PHASE 3 - IMPLEMENT DYNAMIC OVERSIGHT MECHANISM

Theme	Coverage	New	Experts identifiers	Records
P3.1 Situational awareness as a prerequisite	Partial	+	E1, E2, E3, E5	11
P3.2 Explainability aligned with cognitive load	Partial	+	E1, E2, E5	11
P3.3 Monitoring and Intervention Tools	Full		E1, E2, E3, E5	10
P3.4 Bidirectional Interaction - feedback by the users	Full		E1	4
P3.5 Captive vs. voluntary users	None	+	E1, E2, E3	5
P3.6 Mitigating over-trust \ Automation Fatigue	Partial	+	E2, E3, E5	8
P3.7 Adaptive interface design	Partial	+	E1, E2, E4	8
P3.8 Realtime vs. Retrospective decision oversight	None	+	E1, E2	5
P3.9 Controlled switching between LOA	None	+	E3, E5	7
P3.10 Implicit explainability	None	+	E5	2

The thematic analysis of expert interviews revealed a strong emphasis on the need for clear, specific, and actionable definitions of human oversight mechanisms, rather than broad or vague categorizations. This stage of the framework was significantly enriched by novel insights, highlighting the diverse ways in which oversight can and should be operationalized in AI-powered systems. The following themes emerged:

P3.1 Situational awareness as a prerequisite: Effective oversight begins with the user maintaining an understanding of the system's current actions and status, even under high levels of automation. E2 Transcript extract – "*We have to keep the person at some level of awareness so that, if something happens, they can immediately take control*.".

P3.2 Explainability aligned with cognitive load: Explanations should be context-sensitive and aligned with the user's expertise and cognitive capacity, avoiding both oversimplification and unnecessary complexity. E1 Transcript extract – "*You have used all your memory and cannot free anything beyond holding the situation picture in your head. That is one kind of problem. On the other hand, there may be very complicated calculations, which are also cognitive load, but of a completely different kind* ".

P3.5 Captive vs. voluntary users: In environments like military operations, where users are obligated to use the system, oversight can be mandated. In contrast, civilian systems must earn user trust to achieve active oversight. E2 Transcript extract – *" If we turn to people who are not obliged to use it, and the system is very demanding and difficult to handle, they simply will not use it* ".

P3.6 Mitigating over-trust and automation fatigue: Excessive trust in automated systems can lead to disengagement and automation complacency, which is particularly risky in safety-critical settings. Oversight mechanisms should counterbalance this tendency. E3 Transcript extract – "*Being in the loop only passively is very difficult, even just maintaining vigilance. If you want me to look and pay attention, you have to think about how to make me do*. ".

P3.7 Adaptive interface design: Interfaces should be configurable, supporting user-specific alerting, control preferences, and layered transparency, offering simple explanations upfront, with optional access to deeper system logic. E2 Transcript extract – " *As part of training or implementation, we can ask which types of notifications the user would or would not want."*

P3.8 Real-time vs. Retrospective decision oversight: It is crucial to distinguish between decisions that require real-time explanation (e.g., during a mission) and those that can be reviewed post hoc, allowing a balance between operational efficiency and accountability. E2 Transcript extract – " *Some things need to be part of routine operation. Other elements can be placed behind the scenes so that someone can access them calmly, without time constraints*."

P3.9 Controlled switching between levels of automation: The ability to switch between automation levels must be governed by clear, role-based permissions to prevent misuse while supporting flexibility and contextual needs. E3 Transcript extract – "*The question of who can switch control levels is very important, just the user? someone else? a third party?*".

P3.10 Implicit explainability: In many situations, users do not rely on explicit, detailed model-level explanations. Rather, their ability to form shared mental models depends on the system behaving consistently and transparently. When automation becomes unpredictable or opaque, users struggle to understand its actions, weakening their trust and oversight. E5 Transcript extract – "*Even when I show the result without algorithm-level explainability, I still think it has great value and gives the person considerable confidence about what is happening.* "

IV. PHASE 4 - REVEAL CONTRADICTIONS

Theme	Coverage	New	Experts identifiers	Records
P4.1 Oversight vs. Performance	Full		E1, E2	3
P4.2 Transparency vs. Complexity	Partial	+	E1, E2	5
P4.3 Trust vs. Automation	Partial	+	E2, E3	4
P4.4 Performance vs. explainability	Full		E5	2
P4.5 Usability vs. oversight depth	Full		E1, E2	7

P4.2 Transparency vs. Complexity: Providing more detailed information can improve access to the basis of system behavior while increasing the cognitive demands placed on users. Layered explanations can address this tension by presenting concise information during routine interaction and allowing users to access greater detail when needed. E2 Transcript extract – " *Perhaps we should allow different levels ... and direct users to another place that is more complex if they want to go deeper*."

P4.3 Trust vs. Automation: As users develop greater trust in an AI system, they may become more willing to accept higher levels of automation and reduce their direct involvement in decision-making. Conversely, increased automation may encourage passive reliance, reduced vigilance, and skill degradation, potentially diminishing users' ability or readiness to intervene when necessary. E2 transcript extract – "*How much trust the user has in the system can affect how much they want to be involved*.", "E3 explained: "*And the loss of skill, and the distractions (due to automation): I am simply not there".*

V. PHASE 5 - NEGOTIATE TRADE-OFFS

Theme	Coverage	New	Experts identifiers	Records
P5.1 Use of Structured Tools	Full		E4	3

P5.1 Use of Structured Tools such as MCDA, Win-Win negotiation, and stakeholder workshops remain literature-informed candidate methods inherited from Version 1. E4 discusses trade-off deliberation and the limits of explicit weighting, but this does not establish endorsement or effectiveness of the named methods. E5 says that they had not undertaken the proposed structured process. These contextual and qualifying passages are excluded from positive support counts for the specific tools.

VI. PHASE 6 - EVALUATE AND ITERATE

Theme	Coverage	New	Experts identifiers	Records
P6.1 Usability and Trust as Evaluation Metrics	Full		E2	2
P6.2 User-informed evaluation of alternatives	Partial	+	E1, E2, E5	7
P6.3 Objective and subjective UX metrics	Partial	+	E1, E2	9
P6.4 Iterative usability testing	Partial	+	E2	5
P6.5 Post-deployment evaluation	Full		E1, E2	6
P6.6 Need for continual adaptation	Full		E2, E4	3
P6.7 Graduated trust	Partial	+	E1, E2, E4, E5	7

The thematic analysis of the evaluation and iteration phase revealed a strong need to go beyond general methodological references. Participants emphasized the importance of defining clear, specific, and actionable evaluation tasks that align with system goals and user needs. As a result, this phase of the framework was significantly expanded based on the following insights:

P6.2 User-informed evaluation of alternatives: Evaluation should include usability tests and structured feedback to assess how much control or explanation users can realistically manage without compromising system performance. This fosters a negotiation between usability and functional complexity. E2 Transcript extract – " *In these tests we also often combine interviews and try to understand why users did what they did and what they felt interfered with using the system*".

P6.3 Objective and Subjective UX Metrics: Evaluation must incorporate both quantitative (e.g., task success rate, error counts) and qualitative (e.g., perceived ease of use, system usability scores) measures to holistically assess user experience and oversight effectiveness. E2 Transcript extract – " *There are also subjective measures: I ask about satisfaction, how satisfied the person is, and how complex they think the system is."*

P6.4 Iterative Usability Testing: Continuous experimentation, such as A/B testing, scenario-based trials, or interactive prototyping, is essential to determine which oversight mechanisms and interface designs yield the best outcomes. E2 Transcript extract – " *I examined alternatives; I never examined only one way of conveying information, but different ways and compared them*."

P6.7 Graduated Trust: Trust in AI systems develop progressively through repeated exposure to consistent, transparent behavior. Early phases should include stronger oversight, with the option to reduce it as the system proves reliable and users gain confidence. E2 Transcript extract – " *If it makes a mistake at the beginning, it will be much harder for me to continue building trust."*

Appendix - Expert quotations and supported phases

This table presents 96 expert-interview passages, retaining their existing quotation identifiers. English translations reproduce selected extracts, and a passage may inform more than one framework phase or theme. The mapping reflects the authors' interpretation of both explicit expert evaluations and implicit meanings within the experts' observations, examples, and critiques. Explicit evaluations directly address a framework element or oversight concern. Implicit themes and links were developed by interpreting passages in their interview context and triangulating them with the OG-RE4AI conceptual foundation and findings from the earlier stakeholder interviews.

Quote number	Expert ID	English translation of quotation	Supported phase
E1-Q01	E1	It very much depends on the role of the person you want to involve with the system: whether they use the system or receive a service from it.	P1.1 User-role differentiation
E1-Q02	E1	What I am trying to say is that whoever developed the system looks at the different types of users and defines for each of them the level of involvement we are prepared to provide.	P1.1 User-role differentiation P2.1 Clarifying levels of automation
E1-Q03	E1	For the doctor, this is the essence of the job: they have to use the device. They cannot be stuck with a device that says it cannot do something without knowing why.	P1.1 User-role differentiation P3.2 Explainability aligned with cognitive load P3.5 Captive versus voluntary users
E1-Q04	E1	You have used all your memory, and cannot free anything beyond holding the situation picture in your head. That is one kind of problem. On the other hand, there may be very complicated calculations, which are also cognitive load, but of a completely different kind.	P1.5 Environmental constraints P3.1 Situational awareness as a prerequisite P3.2 Explainability aligned with cognitive load
E1-Q05	E1	Situational awareness is a model, a mental model, that people maintain whether or not they understand it; it is simply what happens in the mind.	P3.1 Situational awareness as a prerequisite
E1-Q06	E1	If you take away people's situational-awareness picture, they will not develop trust, they will not have confidence; they will be disconnected from the world.	P1.4 Trust and perception P3.1 Situational awareness as a prerequisite
E1-Q07	E1	This is what connects them to the world. You cannot deprive them of situational awareness; they have to have it. But the smaller processes that build it, anything involving calculation or estimating numbers, can be computerized.	P2.1 Clarifying levels of automation P3.1 Situational awareness as a prerequisite
E1-Q08	E1	They can only say: here it is too short, or here it is too long. They can say qualitative things.	P3.4 Bidirectional interaction and user feedback
E1-Q09	E1	If you first divided this into groups of interfaces, it would be easier to arrive at principles and then compare the groups, because it is very difficult to generalize before understanding the concepts at the lower level.	P1.6 Domain specificity
E1-Q10	E1	Investigation is useful. Following repeated investigations, you can establish a team to examine the traffic-light mechanism in depth because there appears to be a problem and the strategies entered are not suitable.	P6.5 Post-deployment evaluation

Quote number	Expert ID	English translation of quotation	Supported phase
E1-Q11	E1	In this situation, the person does not need it. In another situation, I need to allow deeper investigation and therefore access to deeper levels.	P3.2 Explainability aligned with cognitive load P3.7 Adaptive interface design P4.2 Transparency versus complexity
E1-Q12	E1	When we receive a system with many automatic parts, we want to know that those automatic parts are operating.	P3.3 Monitoring and intervention tools
E1-Q13	E1	When I interact with a human teammate and assign a task, they systematically report that they have started, broken it into parts, are doing this part, will soon do another, and have obtained a result. Your system should report in the same way. When a trainee reports every breath, you tell them they need not report that. The same applies to your system.	P3.3 Monitoring and intervention tools P3.7 Adaptive interface design P6.7 Graduated trust
E1-Q14	E1	Anything outside the here and now is a different process. Once I move into investigation, the rules, goals, everything changes; the person's state of mind changes.	P3.8 Real-time versus retrospective oversight
E1-Q15	E1	In principle, in systems governed by very complex policies, the user generally does not change the policy during operation, but overrides small parts and marks them for later investigation in the appropriate place.	P3.3 Monitoring and intervention tools P3.4 Bidirectional interaction and user feedback P3.8 Real-time versus retrospective oversight P6.5 Post-deployment evaluation
E1-Q16	E1	This is a very important interaction. In real time, I may tolerate the gap between what I would have done and what the system produced. I do not want to override, but I do want to mark it so that I can understand it better later. I do not want that in real time.	P3.4 Bidirectional interaction and user feedback P3.8 Real-time versus retrospective oversight P6.5 Post-deployment evaluation
E1-Q17	E1	Suppose the system says something is suspicious but the doctor disagrees. During the examination, the doctor marks the disagreement with the system at that location, and later it is discussed again. Sometimes the patient is asked to return or have another examination to check the problematic points.	P3.4 Bidirectional interaction and user feedback P6.5 Post-deployment evaluation
E1-Q18	E1	What happens if there is suddenly an error or a malfunction in the system? Does the user know how far back to go to correct it, to change the system's course of behavior to address it, or to identify an impending malfunction, which is more important?	P3.3 Monitoring and intervention tools P6.2 User-informed evaluation of alternatives
E1-Q19	E1	That is what I need; that is my criterion for the system: minimum response time, minimum usage errors.	P4.1 Oversight versus performance P6.3 Objective and subjective UX metrics
E1-Q20	E1	In task analysis with the user, you lay out the process the user performs; at each point, the user says what they expect to receive from the system.	P1.1 User-role differentiation P6.2 User-informed evaluation of alternatives
E1-Q21	E1	Is retrieving the explanation something that task analysis showed is needed in real time, or will the explanation be retrieved in the subsequent investigation?	P3.8 Real-time versus retrospective oversight

Quote number	Expert ID	English translation of quotation	Supported phase
E1-Q22	E1	The more you investigate in real time, the less effective your system will be, because you keep taking side paths outside your main activity.	P4.1 Oversight versus performance P4.5 Usability versus oversight depth
E1-Q23	E1	Ultimately, the work has to get done, rather than someone constantly investigating the system because something does not seem right or asking it to explain what it is doing.	P4.2 Transparency versus complexity P4.5 Usability versus oversight depth
E1-Q24	E1	You really have to be careful here, because otherwise I will end up with a system that is not usable.	P4.5 Usability versus oversight depth
E1-Q25	E1	A system that provides you with a service and a system with which you do your work are completely different; they are two entirely different worlds.	P1.1 User-role differentiation P1.6 Domain specificity P3.5 Captive versus voluntary users
E2-Q01	E2	In battlefield or operational situations, there are influences that are very difficult to predict in the laboratory.	P1.5 Environmental constraints
E2-Q02	E2	We have to keep the person at some level of awareness so that, if something happens, they can immediately take control.	P3.1 Situational awareness as a prerequisite P3.3 Monitoring and intervention tools
E2-Q03	E2	Objective and subjective tests aim to meet particular targets.	P6.3 Objective and subjective UX metrics
E2-Q04	E2	There are also the requirements of the specific domain, or what you call risk management.	P1.2 Domain and risk assessment P1.6 Domain specificity P1.7 Risk factors
E2-Q05	E2	How much trust the user has in the system can affect how much they want to be involved.	P1.4 Trust and perception P4.3 Trust versus automation P6.7 Graduated trust
E2-Q06	E2	If it makes a mistake at the beginning, it will be much harder for me to continue building trust.	P1.4 Trust and perception P6.7 Graduated trust
E2-Q07	E2	That is exactly why we generally aim to be somewhere in the intermediate stages.	P3.7 Adaptive interface design
E2-Q08	E2	It does not matter how reliable the system actually is; what matters is how much the user believes in the system, how reliable they perceive it to be.	P1.4 Trust and perception P3.6 Mitigating over-trust and automation fatigue
E2-Q09	E2	If we turn to people who are not obliged to use it, and the system is very demanding and difficult to handle, they simply will not use it.	P3.5 Captive versus voluntary users P4.5 Usability versus oversight depth
E2-Q10	E2	As part of training or implementation, we can ask which types of notifications the user would or would not want.	P3.3 Monitoring and intervention tools P3.7 Adaptive interface design
E2-Q11	E2	It concerns their understanding: do they really understand what is happening behind the scenes?	P3.1 Situational awareness as a prerequisite P3.2 Explainability aligned with cognitive load

Quote number	Expert ID	English translation of quotation	Supported phase
E2-Q12	E2	It depends on whether you mean why it does it, or what it computes.	P3.2 Explainability aligned with cognitive load
E2-Q13	E2	If you explain it, you need to do so very, very simply.	P3.2 Explainability aligned with cognitive load P4.2 Transparency versus complexity
E2-Q14	E2	Perhaps we should allow different levels, and direct users to another, more complex place if it is important to them to go deeper.	P3.2 Explainability aligned with cognitive load P3.7 Adaptive interface design P4.2 Transparency versus complexity P4.5 Usability versus oversight depth
E2-Q15	E2	I want to know where to put a green light, or that the system decided to put one there based on considerations such as traffic, road width, or an event at a nearby stadium.	P3.1 Situational awareness as a prerequisite P3.2 Explainability aligned with cognitive load
E2-Q16	E2	If someone wants to understand the data or optimization behind the displayed result, the person for whom it is relevant can go and examine it behind the scenes.	P1.1 User-role differentiation P6.2 User-informed evaluation of alternatives
E2-Q17	E2	We want the person not to make mistakes.	P6.4 Iterative usability testing
E2-Q18	E2	Some things need to be part of routine operation. Other elements can be placed behind the scenes so that someone can access them calmly, without time constraints.	P3.8 Real-time versus retrospective oversight P4.1 Oversight versus performance
E2-Q19	E2	One thing about good UX is that everything works; there is nothing to remark on.	P6.3 Objective and subjective UX metrics
E2-Q20	E2	Did the person succeed in performing the task? The second measure is the number of errors, and there is execution speed.	P6.2 User-informed evaluation of alternatives P6.3 Objective and subjective UX metrics
E2-Q21	E2	There are also subjective measures: I ask about satisfaction, how satisfied the person is, and how complex they think the system is.	P6.1 Usability and trust as evaluation metrics P6.3 Objective and subjective UX metrics P6.4 Iterative usability testing
E2-Q22	E2	It is a validated questionnaire with ten statements that the user rates, some phrased positively and some negatively.	P6.3 Objective and subjective UX metrics
E2-Q23	E2	In these tests we also often combine interviews and try to understand why users did what they did and what they felt interfered with using the system.	P6.2 User-informed evaluation of alternatives P6.3 Objective and subjective UX metrics
E2-Q24	E2	Sometimes we set the targets arbitrarily and then update them. We see what is more or less reasonable and examine the alternatives outside.	P4.5 Usability versus oversight depth P6.4 Iterative usability testing P6.6 Need for continual adaptation

Quote number	Expert ID	English translation of quotation	Supported phase
E2-Q25	E2	I examined alternatives; I never examined only one way of conveying information, but different ways and compared them.	P6.2 User-informed evaluation of alternatives P6.4 Iterative usability testing
E2-Q26	E2	Whether completing the process in three minutes is good is a decision we have to make; initially it can be arbitrary, and then we see results and try to understand whether it can be improved.	P6.1 Usability and trust as evaluation metrics P6.3 Objective and subjective UX metrics P6.4 Iterative usability testing P6.6 Need for continual adaptation
E2-Q27	E2	We check how many times the user actually clicks.	P6.5 Post-deployment evaluation
E2-Q28	E2	There are mechanisms that monitor every click on websites, and you can extract that information.	P6.5 Post-deployment evaluation
E2-Q29	E2	If users see that it takes too long, they may stop the process and do it another way.	P4.5 Usability versus oversight depth P6.3 Objective and subjective UX metrics
E2-Q30	E2	I would not necessarily show the ordinary user the whole black box. But I would say: this is your result, this is what is happening now.	P1.1 User-role differentiation P3.2 Explainability aligned with cognitive load P4.2 Transparency versus complexity
E3-Q01	E3	How do we create shared mental models, and what kinds of communication can we use between soldiers, operators, and unmanned systems?	P1.1 User-role differentiation P3.1 Situational awareness as a prerequisite
E3-Q02	E3	We developed a concept for bringing the person into and out of the loop, because that usually happens in emergencies when automation fails.	P3.9 Controlled switching between automation levels
E3-Q03	E3	Precisely in situations where autonomy or automation fails, we should give operators more tools rather than leave them alone.	P3.3 Monitoring and intervention tools P3.9 Controlled switching between automation levels
E3-Q04	E3	The context you have written goes far beyond regulation and risks. It concerns the types of tasks, and the types of tasks greatly influence this allocation.	P1.2 Domain and risk assessment P1.3 Ethical and regulatory constraints P1.7 Risk factors P1.6 Domain specificity P2.1 Clarifying levels of automation
E3-Q05	E3	The range of possibilities for levels of automation is much broader.	P2.1 Clarifying levels of automation
E3-Q06	E3	The intermediate positions can amount to a hundred possibilities. Once you discuss which routines are automated, the main question is what those levels actually are.	P2.1 Clarifying levels of automation
E3-Q07	E3	If we want the person to intervene, they have to understand what their responsibility is.	P1.8 Perceived responsibility and accountability P2.1 Clarifying levels of automation P3.1 Situational awareness as a prerequisite

Quote number	Expert ID	English translation of quotation	Supported phase
E3-Q08	E3	With companion robots for older people, the last thing you want is for the robot to do everything while the person sits still; the robot should encourage movement.	P1.2 Domain and risk assessment P1.6 Domain specificity P2.1 Clarifying levels of automation P3.6 Mitigating over-trust and automation fatigue
E3-Q09	E3	There is also the question of who can move between levels, which is something we are researching now.	P3.9 Controlled switching between automation levels
E3-Q10	E3	It is natural for us that the person decides to step out and let the autopilot operate; the reverse is less natural, when the system decides to do it because it knows better.	P3.6 Mitigating over-trust and automation fatigue P3.9 Controlled switching between automation levels
E3-Q11	E3	There are systems in which the robot is in control, or the technology decides to take command.	P3.9 Controlled switching between automation levels
E3-Q12	E3	The question of who can change the switch is very interesting and very important. Is it only me, another person, or perhaps a third person? There are many questions about who takes responsibility.	P1.1 User-role differentiation P1.8 Perceived responsibility and accountability P3.9 Controlled switching between automation levels
E3-Q13	E3	Being in the loop only passively is very difficult, even just maintaining vigilance. If you want me to look and pay attention, you have to think about how to make me do it.	P1.4 Trust and perception P3.1 Situational awareness as a prerequisite P3.6 Mitigating over-trust and automation fatigue P4.3 Trust versus automation
E3-Q14	E3	And the loss of skill, and the distractions: I am simply not there.	P3.6 Mitigating over-trust and automation fatigue P4.3 Trust versus automation
E3-Q15	E3	Another challenging issue is accountability: when the pilot feels responsible, and how to create conditions in which that responsibility is present.	P1.8 Perceived responsibility and accountability P3.6 Mitigating over-trust and automation fatigue P4.3 Trust versus automation
E3-Q16	E3	What we saw with our military operators was that, regardless of how much we made things difficult for them, they succeeded, because motivation was a factor.	P1.9 User motivation P3.5 Captive versus voluntary users
E3-Q17	E3	Ordinary users either misuse it completely or abandon it, simply abandon it.	P3.5 Captive versus voluntary users
E3-Q18	E3	You want to prevent abandonment; that is why the domain is so important.	P1.2 Domain and risk assessment P1.6 Domain specificity
E3-Q19	E3	The monitoring part is really problematic.	P3.3 Monitoring and intervention tools
E3-Q20	E3	Terminology is very important, because different domains often speak in slightly different terminology.	P1.6 Domain specificity

Quote number	Expert ID	English translation of quotation	Supported phase
E4-Q01	E4	I think that, regarding risks, first I need to understand the level of risk.	P1.2 Domain and risk assessment P1.7 Risk factors
E4-Q02	E4	The interface should communicate it clearly to the user.	P3.7 Adaptive interface design
E4-Q03	E4	I do not think it is possible to be completely comprehensive.	P3.7 Adaptive interface design
E4-Q04	E4	Almost every decision involves a trade-off.	P5.1 Use of structured tools
E4-Q05	E4	Perhaps initially there was more oversight and they gradually loosened it; how do they achieve that balance?	P3.7 Adaptive interface design P6.6 Need for continual adaptation P6.7 Graduated trust
E4-Q06	E4	When you describe what you are looking for, I would not have thought of the word negotiation. I would speak of trade-offs and tools to help you think about and weigh them.	P5.1 Use of structured tools
E4-Q07	E4	Sometimes there are softer things for which it is harder to write the criteria and weight them so clearly.	P5.1 Use of structured tools
E4-Q08	E4	The regulation issue I mentioned at the beginning is, I think, much more significant.	P1.3 Ethical and regulatory constraints
E5-Q01	E5	Your question is very general; you are not focused on systems in transportation or medicine, but looking at things overall.	P1.2 Domain and risk assessment P1.6 Domain specificity
E5-Q02	E5	Explainability goes hand in hand with the level of involvement or trust an operator can place in an autonomous system, especially one based on AI.	P1.4 Trust and perception P3.2 Explainability aligned with cognitive load
E5-Q03	E5	It should be in a language the operator understands.	P1.1 User-role differentiation
E5-Q04	E5	With deep learning it is much harder because explainability is much more limited. Then the operator has to trust you blindly or discard everything you say; there is no gradation.	P1.4 Trust and perception P3.6 Mitigating over-trust and automation fatigue P4.4 Performance versus explainability
E5-Q05	E5	In autonomous navigation, we showed the operator what would happen at the next stage at every moment so that they would gain confidence.	P3.1 Situational awareness as a prerequisite P6.7 Graduated trust
E5-Q06	E5	Even when I show the result without algorithm-level explainability, I still think it has great value and gives the person considerable confidence about what is happening.	P3.2 Explainability aligned with cognitive load P3.10 Implicit explainability
E5-Q07	E5	I think this tension can be addressed through educating the user and building trust by showing results. I do not think this is possible in every domain, but certainly in physical decision-making domains such as navigation.	P1.2 Domain and risk assessment P1.6 Domain specificity P3.10 Implicit explainability
E5-Q08	E5	I asked them initially not to intervene, but to observe what the system recommends and will do. If they thought we were approaching a catastrophe, they should stop it and switch to manual control.	P3.3 Monitoring and intervention tools P3.9 Controlled switching between automation levels P6.2 User-informed evaluation of alternatives

Quote number	Expert ID	English translation of quotation	Supported phase
E5-Q09	E5	That was also how we gained confidence that later it would be possible to operate it without operators.	P6.7 Graduated trust
E5-Q10	E5	It is an intermediate stage needed to build the operator's trust. Later I do not think it is really needed, though for systems making attack decisions, explainability will probably be required over time.	P6.7 Graduated trust
E5-Q11	E5	There are many mechanisms, such as checking eye movements in vehicles, to wake the user, so to speak.	P3.3 Monitoring and intervention tools P3.6 Mitigating over-trust and automation fatigue
E5-Q12	E5	We have not yet started thinking about the operator; we are still developing the algorithms, because regulation there is very strong.	P1.3 Ethical and regulatory constraints
E5-Q13	E5	We try to work with methods in which explainability is much clearer during development, because otherwise it is very difficult.	P4.4 Performance versus explainability